Core technology
Programs

CORE TECHNOLOGY PROGRAMS

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Welcome to the Educational Discoveries Train-The-Trainer program.

Purpose: To train instructors to be competent in presenting and maintaining both the content and context of an Educational Discoveries' designed accelerative learning seminar.

Method: In this program we will use demonstrations, lectures, games, exercises, role play, and presentations, as well as group discussion and feedback.

Results: Participants will learn, practice and demonstrate how to:

- Present an accelerative learning program
- Set a context for learning
- Present the content information
- Connect with yourself, your audience, and your material
- Handle questions

Right Brain/ Left Brain

For some time it has been known that the brain is divided into two sides, left and right. It has also been known that if damage is done to the left side of the brain, the right side of the body tends to become paralyzed and, correspondingly, that if damage is done to the right side of the brain, the left side of the body tends to become paralyzed. In other words, each side of your brain controls the opposite side of your body.

The recent research of Professor Robert Ornstein of the University of California has thrown more light on the different activities handled by each side of the brain. Starting with the realization that the two halves of the brain are biologically similar and can more realistically be thought of as two identical brains working in harmony, rather than as one brain divided into two, Professor Ornstein decided to find out if each of our separate brains handles different intellectual activities in addition to the different physical activities.

His findings were both surprising and significant. In general the left brain handles the following mental activities: mathematics, language, linear and sequential information, logic, analysis, writing, reading, and other similar activities.

The right side of the brain handles very different activities: imagination, color, graphics, music, rhythm, daydreaming, holistic thinking and other similar activities.

Ornstein also found that people who had been trained to use one side of their brain more or less exclusively were relatively unable to use the other side, both in general and in those special situations where the activities specifically related to the other side were particularly needed.

Even more significant, he found that when the "weaker" of the two brains was stimulated and encouraged to work in cooperation with the stronger side, the end result was a great increase in overall ability and effectiveness.

Tony Buzan, *Make The Most of Your Mind*

Notes

Triune Brain

The development of the Triune Brain Model in 1952 has provided great insight into our behavior, ability to learn, and emotional IQ. Dr. Paul MacLean, chief of the Laboratory for Brain Evolution and Behavior at the National Institute for Mental Health in Bethesda, Maryland, created the Triune Brain Model, which delineates three functional brain regions.

MacLean has identified three major layers or "brains," which were established successively in response to evolutionary need. They are the reptilian system, the limbic system, and the neocortex. Each layer is geared toward more or less separate functions, but all three layers interact substantially.

A preliminary way of looking at this is to compare the three layers with three brothers living under one roof. The oldest brother, the reptilian system, is in charge of maintenance. This includes providing food and waste disposal and attending to general security and comfort. Most of his behaviors are automatic and ritualistic, and he strongly resists change. He does not use language, though in some ways he can respond to it.

The second brother, representing the limbic system, feels deeply. He monitors emotion and plays a significant role in remembering new information and organizing events. He is concerned to some extent with self-defense through fight or flight. He also tries to maintain balance between the oldest and the youngest by keeping the oldest brother from totally dominating the family and by serving as a conscience.

The youngest and largest of the brothers is the neocortex. He is creative, can use language, compose music, and engage in very complex analysis. He is capable of formal operational or abstract thought, can anticipate and plan for the future, and is, in general, very intelligent.

Each brother is involved to some extent in the behavior and decisions of the other. They can be supportive of each other. They can also be in conflict. The oldest brothers are most likely to dominate when threat is perceived, because then the top priority is safety and survival. Working together, however, they can successfully deal with quite extraordinary challenges.

Our Ancestral Brain — The Reptilian Complex

The reptilian brain consists largely of the brain stem, the mid-brain, and the reticular activating system. Its purpose is closely related to actual physical survival and overall maintenance of the body. Digestion, reproduction, circulation, breathing, and the execution of the "fight or flight" response in stress are all primarily located in this system. It plays a crucial role in establishing home territory, aggression, reproduction, social dominance, and social hierarchy. Self-preservation, activities of hunting, foraging, hoarding and mating are also centered here. These reptilian behaviors are evident in human beings:

Territoriality

We defend our property both in the physical and abstract sense of ownership, as evidenced by the terms "my house," "my family," "my country," "my room," and "my chair." Note the way that students will pick a seat on the first day of class and automatically sit in that seat for the remainder of the term.

Nesting Behaviors

These are often revealed in the way we fix up the house, decorate it, prepare it for our personal use, and so on. It is, for instance, the place in which we rear our young, so we attach importance to having enough space , storage and supplies, appropriate colors, etc.

Maintaining Social Hierarchies

Despite ostensible commitment to democratic processes, most organizations are based on what have been called "dominance hierarchies." The dominance derives from such factors as perceived biological, intellectual, social or physical superiority.

"Preening" or Ritualistic Display

We do what makes us "stand out" and attract attention, characterized by the sense of "look at me, I am unique."

Flocking Behavior

Flocking is closely linked to social conformity and is found everywhere. In schools we can observe it in groups of students socializing together, perhaps being identified by common hairstyles, clothing, acceptable behaviors determined by "in group" standards, and so forth.

The Guardian at the Gate – The Limbic System

The limbic system is the second and next oldest brain to evolve. It is thought to have developed between 200 to 300 million years ago. It governs the primary centers of emotion – the formation of long term memory, short term memory, conditioned reflexes and attention, as well as metabolism, the immune system and the hormonal system. The limbic system is also involved in primal activities related to food and sex, particularly having to do with our sense of smell and bonding needs, and activities related to expression and mediation of emotions and feelings, including emotions linked to family attachment and bringing up our young. These protective, loving feelings become increasingly complex as the limbic system and the neocortex or "third brain" link up.

The limbic system functions as the regulator of emotions and is also essential to memory. According to MacLean, subjectively, "something doesn't exist unless it's tied up with emotion." He contends that memory is impossible without emotion of some kind, that emotion energizes memory. Personally, we can recognize our facility at remembering events that are emotionally charged. D.A. Rappaport has provided research to indicate that emotions may very well be the medium through which memory patterns are organized. Memory of cognitive material can be heightened when structured within an emotional context or in such a way as to elicit an emotional response.

Greater flexibility of thinking and feeling abilities becomes available through processes that involve the emotions in the learning experience. Emotional repertoire can be expanded through creative stories and assignments that focus on the subtle emotions. Games or

activities that play with emotional context encourage experiences within various emotional contexts. Providing students with the opportunity to express and understand emotions is one of the greatest tools for expanding emotional and intellectual IQ.

The language of expression used by both the limbic and reptilian brain systems is emotive and not linguistic. Current research suggests that the limbic brain regulates much more of our thinking and behavior than we have realized in the past.

The experience of feeling positive about learning is, in itself, a stimulus to encourage excitement about the learning process. When sensory input has a positive emotional reference point, the brain triggers the release of a pleasing opiate type of chemical that creates a state of heightened awareness and well-being.

Within this state, learning becomes pleasurable and is accomplished with greater ease. Physiologically, this can explain the benefits of using music to develop an optimum learning state and assist in memory enhancement.

In the cycle of interaction between the inner and outer worlds, the limbic system represents an essential link between mind and body. We receive as many as a million sensory impressions each minute, but only a select few reach our conscious attention in the neocortex. The limbic system acts as a switchboard, reading the sensations from the body and deciding which to send on to the neocortex for expanded awareness and action. Through this editing process, the limbic system has a significant role in directing our conscious thought.

Our "Thinking" Brain — The Neocortex

Finally, over the limbic brain lies the neocortex, or "thinking cap," the convoluted mass of gray matter that evolved with such rapidity in just the last million years to produce homo sapiens. We share the neocortex with higher mammals such as chimpanzees, dolphins, and whales.

The neocortex constitutes 80% of the total brain matter in the human brain, as well as most of the brain's thinking cortex. It is the outer portion of our brain and is approximately the size of a newspaper page crumpled together. It is divided anatomically into two hemispheres, which are loosely referred to as the left and right hemispheres.

The neocortex makes language, including speech and writing, possible; in that sense, it is different from the other two "brains." Because the neocortex contains about 70% of the brain's approximately 100 billion neurons, it controls high level thought processes such as logic, creative thought, language processing and the integration of sensory information. It renders logical and formal operational thinking possible and allows us to see ahead and plan for the future.

In addition, taking place in these cerebral hemispheres are all the processes concerning vision, hearing, body sensation, intentional motor control, purposeful behavior, and non-verbal ideation. The neocortex includes the frontal lobes of the brain and is also responsible for altruism, compassion, pattern recognition and high level physical/mental coordination.

The prefrontal cortex is that region of the neocortex inside our foreheads. Even though the removal of the prefrontal cortex results in no apparent change in IQ as we now measure it, it appears to be largely responsible for a wide range of abilities that neuro-psychologists call "adaptive behaviors." These include planning, analysis, sequencing, and learning from errors, as well as the inhibition of inappropriate responses and the capacity for abstraction. Thus, increased compassion and empathy become possible because we can acquire the cognitive capacity to "put ourselves in another's shoes."

The creative thought process incorporates the limbic brain with the right and left regions of the neocortex, joining the spark of the "Aha!" illumination with the cognitive and intuitive intelligence of the neocortex. When we strengthen the neural pathways between brain centers, future communications ability strengthens. Brain-center bridging activities have been observed to develop pathways to greater intelligence and thinking potential and even reduce the time required for the brain to make future connections. Using the creative arts to facilitate cognitive instruction provides an opportunity for information to bridge between the logic and intuitive areas of the neocortex.

Under stressful situations, thought processing tends to downshift into the survival-oriented reptilian brain functions or the emotional limbic system. When behavior is limited to these areas alone, significant input from the logical and abstract neocortex and the frontal lobes may be eliminated. Thinking becomes highly limited when access to these brain regions is blocked, and the integration of the abilities and functions of the whole brain cannot occur.

Triune Brain Notes

1. In your own learning experience, what training techniques or aids support the reptilian complex?

2. What training techniques or aids support the limbic system?

3. What training techniques or aids support the neocortex?

Setting the Context

The context in which learning occurs determines its level of success. A context is an environment. It is also a model, and a way of being. It is the most important initial aspect for learning.

In childhood, the *way* a parent interacts with a child determines, to a large degree, the behavior of the child. In a classroom, an instructor's behavior will model the way the learner will respond to openness, connection, humor, spontaneity and flexibility. When an instructor teaches with this context, it is imprinted and transferred to the learner. Those very qualities will make learning fast, easy and permanent.

There are many elements in a positive context for learning, including the physical environment and the emotional attitudes of the instructor and the students. The physical space needs to be bright, well-lit and alive with natural plants, color, artwork, etc. The emotional space is even more powerful in terms of setting context. Whereas it is possible to learn in a bare, stark room, it is not possible for any true learning to occur when fear is present.

Some of the criteria for a context without fear include demonstrating equality among instructors and learners, permission to make mistakes, and valuing all participation, even that which can be deemed disruptive. Equality without fear creates a quality environment where learning can occur

Instructor's Safety Net

Space

Enrollment

Common Ground

Gradient

Ritual

Discovery

Space

Space as a form in presentations is the physical and emotional environment where the presenter and listeners interact. Physical aspects include the temperature, lighting, visuals like color and signage, sound and music, and the seating arrangement for the listeners in the presentation. The emotional space of the room from the perspective of the presenter includes his or her appearance, the manner in which questions are received and responded to, the use of humor, the mode of directions, either command or share, and the openness or closeness of the space.

Enrollment

Enrollment is the process by which people become participants in a program. A typical enrollment includes introducing the instructor and participants, and discussing what's going to be done and why. Enrollment creates the context for learning. Enrollment also gets people interested in doing what you are doing or supporting your idea or action.

By starting with questions that addresses the common experience of the participants, the instructor immediately gets people involved in a physical way. The more multi-sensory and participatory the course activities are, the more learning happens. Note that the questions draw out not only those who are familiar with the topic, but also those who are not. And the last question brings everyone together to agree that their understanding of the subject can use improvement. The instructor's hand is also raised. In that way the instructor and all the participants are treated as equals. These questions set the context of inclusion, participation, and equality.

Common Ground

Participants enter programs with a wide variety of abilities and experience. This is an advantage when the learning activities are group interactive types. And to be effective for everyone, the course must begin at a point where everyone can be included. We call this point "common ground." Common ground refers to the content or experiences shared by participants that give each one a sense of belonging to the group. For example, imagine a seminar on basic accounting with a diverse audience that includes a trainer from Saudi Arabia, a dentist from Denver, and a chief accountant from a major corporation. The seminar starts, and everyone is asked to get comfortable, relax, close their eyes, and listen to, "When you were a child...." Everyone has the common experience of getting comfortable and closing their eyes, and whatever their background, everyone had a childhood. These common experiences, when recalled, together creates for the participants a sense of belonging with the group. Common ground is established.

Gradient

Any mischievous youngster who's ever caught a frog and decided to cook it has learned a lesson about gradient. This lesson is simple–if the water is too hot, the frog will jump out. To cook a live frog, start it in cold water and warm it up slowly. The frog won't notice the increase, and will stay in the pot. The same is true in class: start with low intensity and warm it up slowly. Otherwise the student's attention, like the frog, will leave.

In a learning environment, gradient refers to the level of intensity and commitment required to participate in a given exercise or learning experience. Attending a lecture on the presidential election is low gradient when compared to watching a demonstration on making paper airplanes. A learning segment has higher gradient if the content is more complex or if the personal interaction required is more involved or intimate. Also, when participants are ready for more complex content or more interaction, one can say that they are at a higher gradient. In general, the higher the gradient the greater the learning.

The presenter's personal gradient needs to match the level of the material presented. What emotions, personal information, and interactions with the participants are appropriate, and when?

Ritual

At all major sporting events, the national anthem is played to signify to the crowd that the event is ready to begin. The national anthem has common ground for everyone there. It evokes emotion and creates a communal feeling among all the fans. Creating beginnings and endings, and connecting the participants to each other is the essence of rituals. In program design, a ritual is a formal procedure that, when repeated, connects to deep parts of the brain, especially the reptilian brain. The reptilian brain is where instinctual survival behavior patterns reside. Rituals and habits are in the realm of the reptilian brain.

Discovery

People learn at a deeper level and remember more when they have the opportunity to discover the information instead of just being told. Design and delivery that includes discovery asks more and tells less. Asking "fill-in-the-blanks" type questions increases participation and stimulates creativity. Through participation in exercises designed to evoke emotional experiences, people invariably discover things about themselves and the experience creates new behaviors. Discovery makes information more relevant, personal and memorable.

Notes

Reaction

What Is It?

Reaction is a loss of personal power. It is an unconscious or past response to a present situation. Because we develop unconscious patterns of behavior, we often do not recognize when the outside situation changes. We keep doing things we did for survival and protection, even when they are not appropriate in the present.

Reaction is the way we learned to survive in our families. We developed them initially to cover up those feelings which were unacceptable to express. The way we act to cover those feelings becomes the reaction. When we live in our reactions now, we are creating our lives and our relationships from the past, rather than from the present.

Reaction is also a wonderful opportunity for freedom. If you stay open to learning while in reaction, you can choose a different behavior right in the middle of it. That choice changes the pattern.

You can expect to have the opportunity to experience reaction and change a pattern often during the Train-The-Trainer.

How can you tell if you are "In Reaction"?

A reaction does not have to look a certain way. In fact, it will not. Since it is a learned behavior based on our individual interpretations of how we were supposed to be in our families, a reaction manifests itself differently for everyone. But here are some unmistakable clues that you may be "in reaction":

- when your body feels tight and anxious, not present
- whenever you think that what you feel is about someone else
- when your mind will not shut up
- when your thoughts relate to the past or the future, not the present
- when you are hiding or numb, felling foggy or confused
- whenever there is a judgment (good or bad)
- whenever you feel compelled to act or not act
- whenever you have accumulated good evidence and can prove beyond a shadow of a doubt that you are right
- when events appear to be coming at you – you feel like a victim, a victimizer, and a judge all at the same time

Good Afternoon Ladies and Gentlemen

For you, what was the most difficult emotion to communicate?

What happened instead?

Stop, Look, Listen, Act

The steps for getting out of Reaction:

Stop	**Stop talking, stop thinking, stop doing, just stop.** Take a deep breath. Decide if you want to stay or run. If you decide to run, you are choosing to stay in reaction.
Look	**Look inside.** What are you feeling and why?
Listen	**Listen to your self.** From a calm, centered state, ask yourself, "What do I really want from this interaction?" Independently of the other person(s), how do you want to behave right now when you consider the biggest picture?
Act	Act appropriately.

Notes

The Six Emotional Sources of Questions

Most communication theories agree that words are less than 10% of a communication. Body language, tone, and facial expressions are all considered more important than any literal understanding of words.

There is a well developed, complex language being communicated beneath the words in any interaction. That language is invisible and emotional, and merits as well as demands attention along with words and body language. There are six emotional sources of questions:

How to Handle Questions

Questions from the audience create an opportunity to enroll the whole room, deepen connection, and reinforce learning. They keep the space alive with participation.

When people ask a question, think of it as if they were doing you a favor. Use the F.A.V.O.R. method.

Feel	Feel your feelings. This is your clue to what the questioner is feeling. Put yourself in their place. Determine the emotion that is the source of the question. Notice if you have any judgments about the question.
A - Ask or Acknowledge	Acknowledge the emotion in a non-threatening way by exaggeration, humor, or direct recognition. Ask a question to clarify the real question and confirm that you have received the same emotion that has been sent.
V - Validate	Validate the question and the questioner. This does not mean you have to agree with them. You are letting them know that you've heard them and understand their concern without judgment. Typical responses are "I Understand," "I know what you mean," "I've felt that way myself."
O - Open	Open the space by including the other people in the room. Ask, "Who else has or has had this feeling, experience, or question?" You are giving the questioner time to realize that he/she is not alone. The questioner is also being given time to reexamine the question and its deeper implications.
R - Return	Return to the questioner. Relate your experience, knowledge, or expertise. Let the participant discover the answer to his/her own question.

Notes

Giving and Receiving Support

One of the keys to being successful in the Train-The-trainer and in presenting is being able to give and receive feedback.

Giving Tell the truth of your experience with nothing added or withheld. Do your best not to analyze.

Receiving Welcome everything, no matter how you feel. This is a universal secret and very high gradient. Be compassionate with yourself and the person giving you the feedback. It's as hard for them to give it as it is for you to receive it.

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The Designing Game

For Transforming Training

APPLYING ACCELERATIVE LEARNING TO COURSE DESIGN

DAY 1

DAY 2

APPENDIX